qp: signal alerts

Above: QP.md §3 (signal alerts).

Talks to:

- the desk, whose scheduler starts each cycle and pushes the new signals to the phone (JOBS.md, ALERTS.md §3.6);
- the strategy engine, whose `evaluate()` decides a signal;
- the bar feeds, for live 1-minute bars;
- the shared record, for today's R1 and shortlist;
- qp's own file, where the alert history lives.

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1. What it is

Each saved strategy, watched on today's stocks, and an alert when it fires. Every strategy the trader has saved is evaluated on live 1-minute bars across the union of today's R1 and shortlist, and an alert is raised when a strategy's entry is true on the last completed bar. **It alerts only: nothing is ordered.**

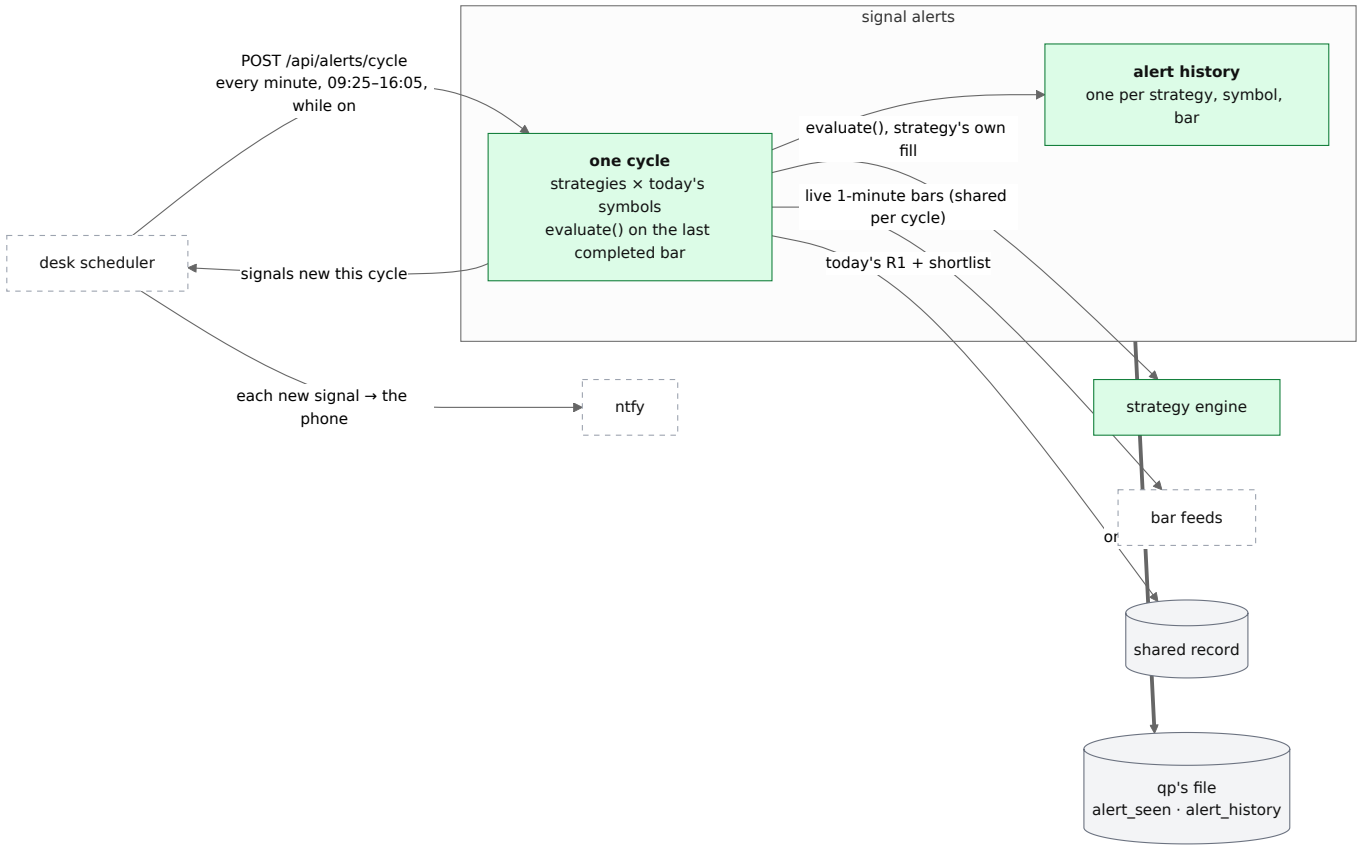
In the design the watcher has **no clock of its own.** The desk's scheduler starts **one cycle** each minute in the session; qp evaluates once, records what it did, and returns the signals that are new in that cycle; the desk pushes each new one to its ntfy topic (ALERTS.md). So a cycle that fails or does not run is a desk job that alerts, and signals reach the phone, not only an open page.

What it owns: the alert history (what fired, when, and on which bar).

What it never does: keep a timer, hold a push key, or place an order.

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2. The diagram



3. Its parts

3.1 One cycle

Point	How
Symbols	the union of today's R1 and shortlist across the tools, read every cycle from the shared record, so a stock flagged at 11:00 is watched from the next cycle
Strategies	every saved strategy, read every cycle: a strategy saved next week is watched automatically. A strategy outside its own window (`window_start` – `window_end` , with 5 minutes of grace before it) is skipped
Signal	the strategy's entry group is true on the **last completed bar**, evaluated with the strategy's own fill model (default `desk`) — the same bar a backtest would trade (Q-N9)
Feed	the desk's default feed, read each cycle (FEEDS.md)
Shared bars	one fetch per symbol per cycle, shared by every strategy on that symbol (not one fetch per strategy-symbol pair)
Isolation	one bad symbol or strategy never stops a cycle; the error is kept in the cycle's record and the other work goes on

3.2 No repeats, across restarts

One alert per strategy, symbol and signal bar. While a signal stays true, repeats are held back for 10 minutes. **The seen-set is stored** (`alert_seen`), so a restart never repeats an alert — today it is held in memory, and a restart may alert once more.

3.3 The history

Every alert is kept (`alert_history`): the strategy, the symbol, the bar, the time, and the cycle it fired in. So alerts can be reviewed, and a restart reads back what already fired.

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4. What changes from today

Change	Why
One cycle per minute, started by the desk's scheduler, not qp's own watcher thread	a cycle that fails or does not run is a recorded desk job that alerts (JOBS.md); today qp runs its own thread and a dead thread is silent
Signals pushed to the phone by the desk	today a signal reaches only an open page (a sound and a browser notification); qp holds no push key
The strategy's own fill model (default `desk`), not `close`	an alert and a backtest of the same strategy must take the same bar (Q-N9)
Alert history and the seen-set stored	a restart never repeats an alert, and alerts can be reviewed
Shared bars per cycle	one fetch per symbol, not per strategy-symbol pair
The default feed, read each cycle	it was fixed to Polygon, a day behind on the free plan (F-22)

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5. Contracts

The code: `quant-platform/chart/alerts.py` .

It owns (qp's file, `~/.qp/platform.db`)	Key
`alert_seen` (the suppression set)	strategy, symbol, bar
`alert_history` (what fired)	strategy, symbol, bar

It reads	From
today's R1 and shortlist	the shared record
live 1-minute bars	the bar feeds (FEEDS.md)
every saved strategy	qp's own file (STRATEGIES.md)

Route, behind `/quant/`	Does
`POST /api/alerts/cycle` (**to build**)	run one cycle, started by the desk's scheduler; answer with the signals new in that cycle
`GET /api/alerts/status` , `…/recent`	state, what it watches, last cycle, errors; alerts newer than a time (an open page still polls this)

The old `POST /api/alerts/start` / `stop` (qp's own watcher) retire with the switch to the desk's scheduler.

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6. When it fails

Event	What happens
qp is down	each cycle the scheduler starts fails, is recorded and alerts (JOBS.md)
one strategy or symbol throws in a cycle	the cycle goes on with the others; the error is in the cycle's record
a feed returns nothing for a symbol	that symbol is skipped this cycle and counted; no false signal
the shared record is down	the cycle has no symbol list and says so; it alerts as a failed job
qp restarts	the seen-set is read back from the file, so no alert repeats

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7. Built on

The old desk: the live signal watcher in legacy/qp/ALERTS-AND-PRINTS.md.

What failed there:

ID	Fault	Prevented by
Q-N9	the watcher evaluates with `close` while backtests default to `desk`, so an alert can fire on a different bar	the strategy's own fill model, the same as the backtest
F-22	the watcher was fixed to Polygon (a day behind on the free plan)	the desk's default feed, read each cycle
—	the watcher is qp's own thread; a dead thread is silent	one cycle per minute, a recorded desk job
—	a signal reached only an open page	the desk pushes each new signal to the phone
—	alert repeats were held in memory; a restart could re-alert	the seen-set and history stored

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8. Status

Part	Status
alerts on each strategy's own fill model; alert history kept	**built, live** (step 12)
cycles started by the desk's scheduler, new signals pushed to the phone	**to build** (step C) — today qp runs its own watcher thread and signals reach only an open page
today's lists read from the shared record	moves with the register switch (step D; BACKTEST.md)